Company: Clonify (clonify.ca)

Clonify Microbots — Generative Physical Matter



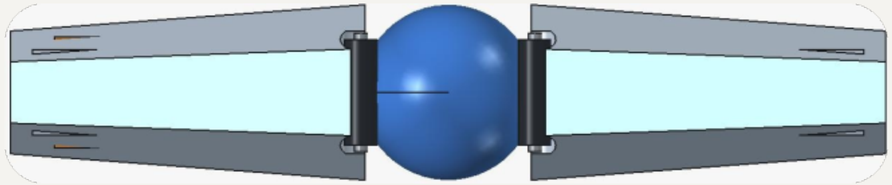
Founder & Contact: Noah Abebe (mail@clonify.ca) | Toronto, ON

Image Illustrative for decorations [and] not descriptive of the product

The Vision

Prompt to Physical Reality

Moving beyond digital screens into software-defined, dynamic hardware. We are bringing cinematic technology to life to enable seamless, real-time physical assembly.

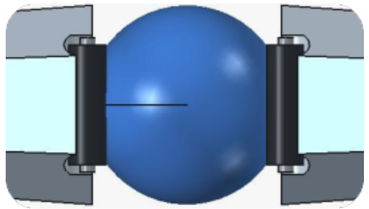


The Problem: Static manufacturing takes months to prototype new physical shapes, limiting rapid iterations in robotics and rapid tooling.

The Solution: A swarm of modular, reconfigurable microbots that transform text or CAD prompts directly into physical structures in real time.

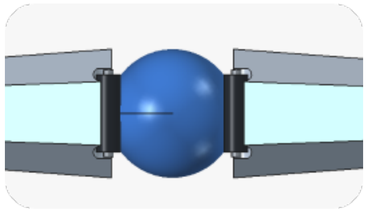
System Design

Product Architecture (The CAD Model)



Magnetic Mesh Networking

Integrates EPM cores for zero-power stability with mechanical interlocking joints. Contact pads enable high-bandwidth inter-bot power and data mesh.



Kinetic CAD Translation

Converts digital 3D meshes into physical actuation instructions. The primary software interface bridging CAD design intent with mechanical bot movement.



Swarm Behavior Engine

AI pipeline parsing spatial voxel grids into collision-free paths. Manages the emergent swarm logic required for complex structural assembly.

Opportunity

Market & Initial Applications

- Educational Robotics
- Research Labs
- Emergency Shoring
- Space Hardware

Serviceable Addressable Market

$1.8B

Research Robotics & Modular Automation Market

Execution Roadmap

6-Month IoT Compliance & Security Plan

1

Months 1–2: Device Security & Hardware

Establishment of Hardware Root of Trust (RoT) and Secure Boot protocols alongside high-resolution SLA/SLS chassis printing and resin optimization.

2

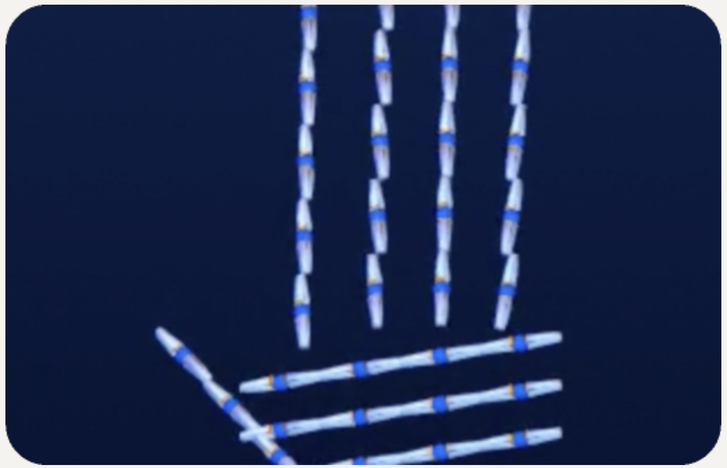
Months 3–4: Connectivity & Data Management

PCB integration of IR transceivers with encrypted communication (TLS/MQTTS) and localized data management silos for IoT compliance.

3

Months 5–6: Lifecycle & Swarm Deployment

100-unit swarm deployment running prompt-to-shape algorithms with secure Over-the-Air (OTA) updates and automated decommissioning protocols.



Leadership

The Founder & Team

Noah Abebe

Founder



Computer Science (York University), Diploma in Computer Programming (Seneca).

Expertise in full-stack architecture, AI/ML evaluation, and FIRST/WRO robotics coaching.

Pre-Seed Ask

Funding Ask & Budget Allocation

$350,000 CAD

Expense Category	Allocation	Purpose
R&D & Embedded Engineering	$140,000	Firmware development, control loops, and AI spatial solver
3D Printing & Tooling	$70,000	Industrial SLA/SLS carbon-nylon iterations (100+ units)
Electronics & Custom Coils	$60,000	Custom PCBs, EPM drivers, neodymium components
IP & Patent Filing	$30,000	Provisional utility patents for locking mechanism
Operations & Facilities	$35,000	Testing equipment, tracking camera array

Market Pain Point

The Industrial Problem

Heavy industrial tooling, temporary scaffolding, and rescue structures are rigid, slow to deploy, and single-purpose.

Industry requires dynamic, self-assembling materials capable of high shear and tensile load tolerance.



Software-Defined Reality

Layer 01: Intent

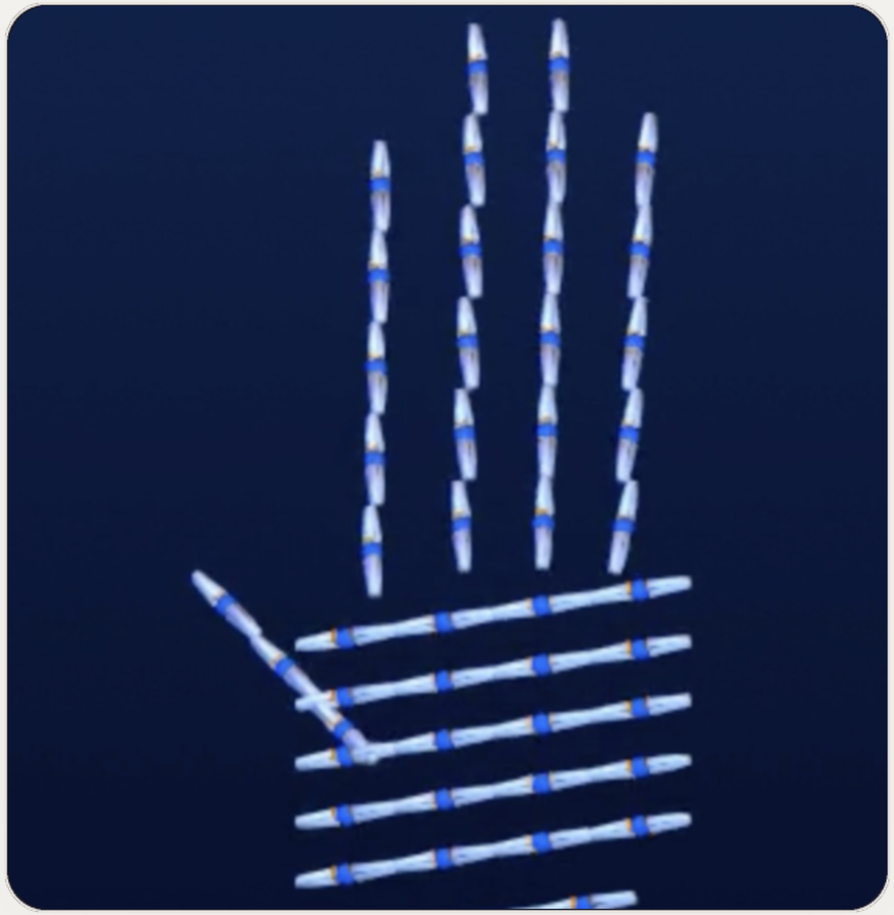
User prompts are parsed into spatial voxel data, defining the target geometry.

Layer 02: Swarm OS

Real-time FEA ensures structural integrity while the orchestration engine assigns paths to individual modules.

Layer 03: Physical Execution

Modules dock and lock, translating software code into a tangible, high-strength physical object.



Commercialization

Business & Licensing Model

B2B Hardware Sales

High-margin industrial swarm node units for infrastructure and defense.

Software Licensing

Recurring enterprise license for the Spatial AI Pathfinding Engine.

